INCH-POUND
MIL-DTL-40626E
18 April 2025
SUPERSEDING
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31 March 1994

DETAIL SPECIFICATION

SIGN KIT, VEHICLE CLASS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers one type and size of vehicle class sign kit.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-S-698	Steel, Sheet and Strip, Low Carbon
TT-C-490	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)
TT-P-1757	Primer Coating, ALKYD Base, One Component

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AMSC N/A FSC 9905

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-53039	Camouflage Coating. Solvent Based, Chemical Agent Resistant

(Copies of these documents are available online at https://assist.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-2248 Sign kit, Vehicle Class; Assembly and Detail

(Copies of drawing are available from US Army Soldier Systems Center, www.army.mil/natick.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

ASTM INTERNATIONAL

ASTM B209/B209M	Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B211/B211M	Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE-AMS-QQ-A-250/8	Aluminum Alloy 5052, Plate and Sheet

(Copies of these documents are available online at www.sae.org or SAE International Customer

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Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material and components. The materials and components shall be as specified herein and as shown on the applicable drawing. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Aluminum.

3.2.1.1 Plate and sheet. The aluminum alloy plate and sheet shall be Alloy 5052 of ASTM B209/B209M or as specified in SAE AMS-QQ-A-250/8.

3.2.1.2 Bar, rod, and wire. The aluminum alloy bar, rod, and wire shall be Alloy 2011, temper T3 of ASTM B211/B211M.

3.2.2 Steel. Sheet, and strip. The steel sheet shall conform to cold rolled or hot rolled, commercial quality, and the steel strip shall conform to any temper, condition and edge of QQ-S-698.

3.2.3 Cap screws, washers, and nuts. Cap screws, washers and nuts shall conform to the requirements shown on Drawing 6-1-2248.

3.2.4 Primer. The primer shall be as specified in TT-P-1757 with composition and color optional.

3.2.5 Coating. The coating shall be as specified in MIL-DTL-53039.

3.3 Construction. The sign kit shall conform in all respects to Drawing 6-1-2248 and as specified herein.

3.3.1 Disk. The disk shall be fabricated of aluminum alloy, temper 0, as specified in 3.2.1.1.

3.3.2 Anchor pin and pin lock. The anchor pin and pin lock shall be fabricated of aluminum alloy specified in 3.2.1.2. The pins after assembly, shall not be loose.

3.3.3 Plates, retaining and numeral. The retaining and numeral plates shall be fabricated of aluminum alloy, temper H38, as specified in 3.2.1.1. The numeral and retaining plates shall be inserted in position and shall not become loose or fall out due to vibration when tested as

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specified in 4.3.5. Prepainted aluminum sheet for numeral plates may be furnished and shall conform to the manufacturers' standard commercial practice. The background color shall be Dark Green 34082, as specified in 3.2.5.

3.3.4 Brackets, universal, long and short. The brackets shall be fabricated of steel as specified in 3.2.2 and shall be of the design as shown on Drawing 6-1-2248.

3.3 Finish. Except when prepainted sheet is used, the sign kit shall be finished as specified herein.

3.4.1 Surface preparation. All surfaces and edges of the disk, retaining plate, and numeral plates shall be prepared for painting in accordance with class 1A of MIL-DTL-5541. The brackets shall be prepared for painting in accordance with any method of TT-C-490.

3.4.2 Priming. All prepared surfaces and edges of disks, retaining plate, and numeral plates shall be coated with the primer specified in 3.2.4.

3.4.3 Coating. After drying, the primed surfaces and edges of the disk, retaining plate, steel brackets, and numeral plates shall be coated as specified in 3.2.5.

3.4.4 Characters. The letter "C" and the numerals shall be gothic type and shall be applied to the retaining plate and numeral plates, respectively, after the background color specified in 3.3.3 has been applied. The letter "C" and the numerals shall be clear and sharply defined. Additional requirements for the characters shall be specified in Table I.

3.5 Marking for identification. Each sign kit disk shall be permanently and legibly marked on the back side with the manufacturer's name or readily identifiable trademark.

TABLE I. Additional character requirements.

Requirement	Letter	Numeral	
Height	2 inch minimum 2.625 inch minimum	3.500 inch minimum 3.625 inch minimum	
Width of stroke	0.438 inch minimum 0.563 inch minimum	0.563 inch minimum 0.688 inch minimum	
Color (see 3.2.5)	Aircraft Red 31136	Black 37030	
Data	Letter "C" centered on one side of the retaining plate only	One side 0 2 4 6 8	Opposite side 1 3 5 7 9

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3.6 Workmanship. The end item shall conform to the quality of product established by this specification. All edges shall be rounded or beveled, and all surfaces and edges shall be free from burrs, silver and sharp edges. The finish of the disks, retaining plate, numeral plates, brackets, lettering and numerals shall be free from imbedded foreign matter, uncoated areas, cracks, blisters, flakes, scratches, mars, sags and runs. The lettering shall be smooth, even, straight and perpendicular to the base.

4 VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.3.3 and 4.3.4 and tested for the characteristics specified in 4.4.1. Any nonconformance shall be cause for the rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ Z1.4.

4.3.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In process inspection., Whan applicable, inspection shall be made to determine that surface preparation and application of primer to the components are in accordance with the specified requirements of 3.4.1 and 3.4.2. Whenever nonconformance is noted, correction shall be made to the affected items and process. Items that cannot be corrected shall be removed from the process.

4.3.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be I.

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TABLE II. End item visual defects.

Examine	Defect	Classification Major	Minor
Finish (all components)	Component not colors specified	101	
	Cap screws, nuts and washers not zinc plated		201
	Coated surfaces not smooth, uniform or continuous		202
	Uncoated areas	102	
	Not free from cracks, blisters, orange peel, flakes, sags, runs, scratches or mars		203
	Foreign matter imbedded		204
	Touch up, when applicable, not uniformly accomplished		205
Design	Component parts do not assemble together or fit snugly	103	
Construction and workmanship	Any hole missing	104	
	Embossments missing or not located as shown on drawing	105	
	Bracket and disk holes not in alignment	106	
	Pins not securely riveted	107	
	Rivet heads not counter sunk as shown		206
	Characters not gothic type	108	
	Letter “C” not applied as specified		207
	Numerals not applied on both surfaces of numeral plates	109	
	Numeral plates not numbered as specified	110	
	Not required number of numeral plates	111	
	Characters not centered		208
	Sharp edges, burrs or slivers not removed	112	
	Edges and corners not rounded		209
	Edge of letter or numerals not clearly defined		210
Identification marking	Missing, incomplete, not legible		211

4.3.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 6-1-2248 and as specified herein. Only those dimensions that can be evaluated without damaging or disassembling the end item shall be examined. Any dimensions not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be S-4.

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4.3.5 End item testing. The end items shall be tested for the vibration characteristic listed in 4.4.1. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be S-3.

4.4 Methods of inspection.

4.4.1 Vibration test. The sign kit assembly, with brackets attached to simulate mounting on a vehicle in an upright position, shall be affixed to any commercial vibration testing machine, and tested for compliance with 3.3.3. The vibration test shall be conducted for one hour at a frequency rate of 1000 cycles per minute at an amplitude of 0.25 inch. Any numeral, letter, or retaining plate falling out due to vibration or found to be loose after the vibration test shall constitute failure of this test.

5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vehicle class sign kit is used to afford immediate recognition of the weight classification of each vehicle and as an aid in traffic control at bridges and crossings.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.2 and 6.3).
c. Packaging requirements. (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement as cited in NATO STANAG No. 2021. When amendment, revision, or cancellation of this specification is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization office, if required.

6.5 Subject term (key word) listing.

Identification marking
Traffic control
Vehicle weight
Weight classification

6.6 Change from previous issue. Marginal notations are not used in revision to identify changes with respect to the previous issue due the extensiveness of the changes.

Custodians:
Army - GL
Navy - YD
DLA - CQ

Preparing activity:
DLA - CQ

(Project 9905-2025-002)

Review activities:
Army - AT,CR4, MD
Navy - MC

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